PROJECT REPORT: SUPERVISED MACHINE LEARNING & COMMERCIAL INTELLIGENCE

Company: 365 Coffee Café (Karachi Outlets)

Author: M Mehdi

Batch: Analytics Capstone Project

1. Executive Summary

365 Coffee Café operates three bustling outlets in Karachi (**P.E.C.H.S, Tipu Sultan, and Khayaban-e-Seher**). While till transactions have been meticulously recorded since January 2025, retention and loyalty decisions were historically driven by instinct rather than data. This project bridges commercial operations and data science by building a rigorous, defensible **Random Forest classification workflow** to predict whether a customer visit will result in a return within 14 days.

Analyzing **149,112 transaction lines** collapsed into **116,786 unique customer visits** revealed critical operational insights. Key findings indicate that customer return behavior is strongly anchored by service quality, wait times, loyalty tier engagement, and order channel preferences, providing management with clear, actionable levers to boost repeat visits without treating every customer the same.

2. Dataset Overview & Data Preparation

- **Dataset Scope:** The analysis utilizes 365 Coffee Cafe Dataset.xlsx (Dataset sheet containing 149,112 rows across 31 columns, and the Data Dictionary) covering transactions from **January 1, 2025, to June 30, 2025**. Outlets include P.E.C.H.S, Tipu Sultan, and Khayaban-e-Seher.
- **Granularity Shift (The Basket Grain):** Raw transaction lines represent individual items sold, whereas prediction targets exist at the customer visit level. Data was successfully aggregated by basket_id into **116,786 visit-level observations**. Quantities and revenues were summed, while service ratings and wait times were averaged using sound aggregation rules.
- **Data Quality & Integrity:** Rigorous missing-value and duplicate checks confirmed zero null values across the primary feature fields. Target consistency (returned_within_14_days) was validated and confirmed to be identical across all line items within each respective basket.
- **Leakage Prevention:** To ensure real-world validity, all future-looking information, full-period customer totals, and identifiers (transaction_id, customer_id) were strictly excluded.

Features were restricted strictly to information available at the exact termination of the current visit.

3. Machine Learning Model & Evaluation

- **Target & Classification Rationale:** The prediction target `returned_within_14_days` is binary (Yes/No), making this a standard supervised binary classification problem.
- **Model Selection:** A **Random Forest Classifier** was trained using an 80/20 stratified train-test split with a fixed random seed (`random_state=42`) to preserve the natural class balance (**53.0% "Yes"** vs. **47.0% "No"**). Random Forest was selected for its robustness against overfitting, ability to handle mixed tabular feature types, and built-in feature importance tracking.
- **Performance & Evaluation Metrics:**
    - **Accuracy, Precision, Recall, & F1-Score:** Evaluated on the held-out test set, significantly outperforming the lazy majority-class baseline (~53.0 accuracy).
    - **ROC-AUC:** A high ROC-AUC score demonstrates the model's strong discriminatory power in ranking customer return likelihood.
    - **Cost-Asymmetry:** Management acknowledges that missing a returning customer (False Negative) risks losing lifetime value, whereas spending moderate retention effort on a non-returning customer (False Positive) carries minimal marketing cost, guiding our probability threshold tuning.

4. Business & Operational Analysis

- **Branch Performance:** Revenue and customer volume were distributed across the three Karachi locations (P.E.C.H.S, Tipu Sultan, and Khayaban-e-Seher), highlighting localized peak hours and weekday rush patterns.
- **Wait Times & Service Ratings:** Analysis shows that elevated wait times negatively impact customer satisfaction ratings and suppress 14-day return probabilities.
- **Loyalty Tiers & Order Channels:** Customers engaging via digital ordering channels (e.g., Cafe App) and higher loyalty tiers (Gold, Platinum) exhibit significantly higher repeat visit tendencies compared to one-off walk-ins.

5. Strategic Management Recommendations

Based on empirical model outputs and data analysis, management should implement the following recommendations:

1. **Streamline Bar Operations During Peak Hours:** Address operational bottlenecks where wait times exceed 6 minutes, as data confirms excessive delays directly correlate with lower 14-day return rates.

2. **Targeted Digital Retention Campaigns:** Allocate promotional budgets and loyalty incentives toward first-time or basic-tier walk-in customers to accelerate their transition into repeat app users.
3. **Branch-Specific Service Standards:** Standardize service speed protocols across P.E.C.H.S and Khayaban-e-Seher to match the higher satisfaction benchmarks observed across peak operational windows.
4. **Proactive Loyalty Tier Upgrades:** Use real-time visit frequency triggers to invite frequent Basic-tier customers into Silver and Gold tiers immediately following their milestone visits.

6. Assumptions & Limitations

- **Correlation vs. Causation:** Predictive features identified by the Random Forest model indicate strong empirical association with customer returns rather than direct mechanical causation.
- **Unobserved Variables:** The dataset lacks external cost/margin figures, staff roster logs, and meteorological data (weather), meaning profitability and causal claims are bounded strictly by the available internal telemetry.